

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Inorganic Cation Transmembrane Transport	0.08675135	235	5.324e-06	1.614e-02	ANO6:95 SLC6A12:119 SLC36A2:138 HPN1:160 SLC6A15:194 ATP6V0A4:196
Monatomic Cation Transmembrane Transport	0.08536139	234	7.845e-06	1.614e-02	ANO6:95 SLC6A12:119 SLC36A2:138 HPN1:160 SLC6A15:194 ATP6V0A4:196
Monocarboxylic Acid Transport (GO:001571)	0.18171165	50	0.015e-06	1.614e-02	SLC5A8:64 SLC5A12:67 SLC6A12:119 ABCD3:167 FABP1:179 KDM5A:401
Organic Hydroxy Compound Transport (GO:0	0.20888495	36	1.468e-05	1.971e-02	SLC22A1:62 SLC5A8:64 ABCD3:167 GHRL:252 SLC51B:455 SLC51A:698
Carboxylic Acid Transport (GO:0046942)	0.16184370	47	1.259e-04	1.353e-01	SLC5A12:67 SLC6A12:119 SLC36A2:138 SLC6A15:194 SLC6A18:429 SLC6A6:478
Intracellular Monoatomic Ion Homeostasis	0.23354165	19	4.269e-04	3.822e-01	CLDN16:103 CNNM4:204 SLC4A1:312 LETM1:510 SLC39A10:812 RHCG:813
Organic Anion Transport (GO:0015711)	0.11847502	72	5.213e-04	4.000e-01	SLC05A1:111 SLC52A3:143 SLC4A9:176 SLC4A1:312 SLC4A3:390 MGST1:402
Sodium Ion Transmembrane Transport (GO:0	0.11890770	67	7.794e-04	5.232e-01	SLC6A12:119 SLC6A15:194 SLC9A4:319 SLC22A1:62 SLC5A8:64 SCNN1G:358 SCNN1B:406
Organic Cation Transport (GO:0015695)	0.12707857	30	9.747e-04	5.817e-01	SLC22A1:62 SLC6A12:119 SLC18B1:126 SLC22A4:675 SLC5A7:766 SLC44A5:769
Bile Acid And Bile Salt Transport (GO:00	0.23220741	16	1.305e-03	5.843e-01	ABCD3:167 SLC51B:455 SLC51A:698 SLC10A2:869 SLC10A3:891 SLC27A5:1144
Nitrogen Compound Transport (GO:0071705)	0.08104306	133	1.301e-03	5.843e-01	SLC22A1:62 SLC5A8:64 SLC6A12:119 SLC36A2:138 SLC52A3:143 SLC6A15:194
Sulfur Compound Transport (GO:0072348)	0.20871940	20	1.238e-03	5.843e-01	SLC22A1:62 SLC26A4:296 SLC26A8:768 SLC13A1:798 SLC35B3:826 SLC25A10:848
L-phenylalanine Catabolic Process (GO:00	0.42004141	4	3.621e-03	6.145e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA
L-phenylalanine Metabolic Process (GO:00	0.42004141	4	3.621e-03	6.145e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA
Acid Secretion (GO:0046717)	0.31814180	7	3.561e-03	6.145e-01	DRD3:243 GHRL:252 SLC51B:455 SLC1A1:698 SLC26A6:3075 SLC9A3R1:3295
Alpha-Amino Acid Biosynthetic Process (G	0.27832458	10	2.310e-03	6.145e-01	BCAT2:28 SDSL:134 CAD:730 SDS:847 CPSI:1137 BCAT1:2925
Canonical Wnt Signaling Pathway (GO:0060	-0.15408568	33	2.066e-03	6.145e-01	PTEN:5 WNT4:25 DVL2:140 FRAT1:358 FZD6:407 BCL9L:411
Erythrose 4-Phosphate/Phosphoenolpyruvat	0.42004141	4	3.621e-03	6.145e-01	TAT:546 GSTZ1:673 HGD:1024 IL4I1:1697 NA NA
Ethanol Metabolic Process (GO:0006067)	0.34505051	6	3.425e-03	6.145e-01	ACSS2:412 ADH4:1182 SULT1A2:1487 SULT1E1:2135 SULT1A3:2950 SULT1B1:3463
Gamma-Aminobutyric Acid Transport (GO:00	0.38776891	5	2.676e-03	6.145e-01	SLC6A12:119 SLC6A6:478 SLC6A11:646 SLC2A13:2492 SLC9A3R1:3295 NA
Innate Immune Response In Mucosa (GO:000	0.42198546	4	3.468e-03	6.145e-01	BPIFB1:2 NOS2:197 PLA2G1B:225 IL4:3480 NA NA
Inorganic Cation Import Across Plasma Mem	0.09640985	85	2.172e-03	6.145e-01	SLC12A9:271 CLU:299 SLC12A6:305 SLC9A4:319 SLC12A3:356 SCNN1G:358
Monoatomic Cation Homeostasis (GO:005508	0.11232214	61	2.450e-03	6.145e-01	CLDN16:103 CNNM4:204 SLC12A9:271 SLC12A6:305 SLC12A3:356 SCNN1G:358
Mucosal Immune Response (GO:002385)	0.28314470	10	1.935e-03	6.145e-01	BPIFB1:2 NOS2:197 PLA2G1B:225 FFAR2:426 IFNLR1:1338 RAB17:2714
Negative Regulation Of Gene Expression V	0.37637612	5	3.562e-03	6.145e-01	ZNF445:338 UHRF1:1311 USP7:1572 HELLS:1669 UHRF2:2788 NA
Negative Regulation Of miRNA Metabolic P	0.21677282	15	3.661e-03	6.145e-01	POU5F1:270 PDGFB:603 ESR1:1493 HIF1A:1560 NFATC4:1662 NFATC3:1690
Negative Regulation Of mRNA Transcriptio	0.24167076	14	1.748e-03	6.145e-01	POU5F1:270 PDGFB:603 ESR1:1493 HIF1A:1560 NFATC4:1662 NFATC3:1690
Negative Regulation Of Single Stranded V	0.34856061	6	3.111e-03	6.145e-01	TASOR:5 MORC2:30 INPP5K:534 SETDB1:1016 MPHOSPH8:1841 HMG2A:8031.5
Positive Regulation Of DNA Methylation-D	0.23202278	8	1.782e-03	6.145e-01	TASOR:5 MORC2:30 SETDB1:1016 TRIM28:1724 MPHOSPH8:1841 PPHLN1:2236
Positive Regulation Of Heterochromatin F	0.28643169	10	3.294e-03	6.145e-01	TASOR:5 MORC2:30 SETDB1:1016 TRIM28:1724 MPHOSPH8:1841 PPHLN1:2236
Regulation Of Cell Adhesion (GO:0030155)	-0.08285646	105	3.441e-03	6.145e-01	TGM2:44 PAHB:70 RAC2:161 CXCR3:394 FUT4:400 FES:402
Sodium Ion Transport (GO:0006814)	0.10041948	74	2.872e-03	6.145e-01	SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC6A15:194 SLC12A3:356 SCNN1G:358
One-Carbon Compound Transport (GO:001975	0.15367119	29	2.056e-03	6.643e-01	AQP11:35 SLC4A9:176 SLC4A1:312 SLC4A3:390 RHAG:855 CA2:947
Sulfate Transport (GO:0008272)	0.26202506	10	4.121e-03	6.643e-01	SLC26A4:296 SLC26A8:768 SLC13A1:798 SLC25A10:848 SLC26A2:2397 SLC13A4:2561
Chromosome Condensation (GO:0030261)	0.18648092	19	4.909e-03	7.202e-01	HI-6:139 AKAP8L:527 PLK1:811 AIFM1:853 ERN2:964 NCAPH:1191
Fat-Soluble Vitamin Metabolic Process (G	0.20373483	16	4.793e-03	7.202e-01	CYP11A1:4 LRAT:112 NQO1:374 KDM5A:401 GGX5:505 CYP1A1:665
Intracellular pH Elevation (GO:0051454)	0.30573074	7	5.095e-03	7.202e-01	NOX1:213 TMEM175:657 CLN3:1383 SLC26A3:1580 OCA2:2165 SLC26A6:3075
Replication Fork Processing (GO:0031297)	-0.13343788	37	4.995e-03	7.202e-01	BRCA1:247 RAD51:284 MUS81:369 PARP1:4699.5 ATRX:4699.5 WRN:4699.5
Aromatic Amino Acid Family Catabolic Pro	0.26303706	9	6.292e-03	8.242e-01	TAT:546 GSTZ1:673 HPD:760 HGD:1024 IL4I1:1697 MKO:2841
Sodium Ion Homeostasis (GO:0055078)	0.19151404	17	6.278e-03	8.242e-01	SLC12A3:356 SCNN1G:358 SCNN1B:406 ATP1B3:477 SCNN1A:653 ATP4B:1118

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.12279528	219	4.404e-01	2.852e-06	SLC1A1:12 SLC22A1:62 SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC36A2:138
REACTOME_TRANSPORT_OF_SMALL_MOLECULES	0.06514096	566	1.566e-07	3.382e-04	SLC1A1:12 AQP11:35 MTPP:36 ALB:51 SLC22A1:62 SLC5A8:64
WP_NRF2_PATHWAY	0.16050922	92	1.077e-07	3.382e-04	TXNRD3:58 SLC5A8:64 SLC5A12:67 MGST3:80 SLC2A9:173 SLC6A15:194
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS	0.14702944	95	7.612e-07	1.232e-03	SLC1A1:12 SLC5A8:64 SLC5A12:67 SLC6A12:119 SLC36A2:138 SLC4A9:176
WP_NUCLEAR_RECEPTORS_METAPATHWAY	0.09475113	222	1.257e-06	1.628e-03	EHHADH:29 ABCB11:56 TXNRD3:58 SLC5A8:64 SLC5A12:67 MGST3:80
REACTOME_TRANSPORT_OF_BILE_SALTS_AND_ORG	0.15728318	77	1.880e-06	1.951e-03	SLC22A1:62 SLC6A12:119 SLC6A15:194 SLC13A2:205 SLC6A18:429 SLC6A6:478
REACTOME_SLC_TRANSPORTER_DISORDERS	0.14901738	85	2.109e-06	1.951e-03	SLC1A1:12 SLC36A2:138 NUP133:147 SLC2A9:173 SLC26A4:296 SLC12A6:305
CAIRO_HEPATOBLASTOMA_DN	0.09447100	196	5.490e-06	4.444e-03	SLC1A1:12 GYS2:34 ABCB11:56 SLC22A1:62 GCDH:78 HP:90
WP_TRANSULFURATION_ONECARBON_METABOLISM	0.18126636	51	7.645e-06	5.501e-03	BCAT2:28 CHDH:127 GNMT:174 PHGDH:301 GPX4:371 TYMS:468
FEVR_CTTNMB1_TARGETS_UP	0.05261648	502	6.327e-05	3.854e-02	CLPS:1 SLC1A1:12 ENTDP5:20 SPTA1:33 AQP11:35 MTPP:36
LEE_LIVER_CANCER_DENA_DN	0.14481559	63	7.142e-05	3.854e-02	EHHADH:29 PNLP1:37 GCDH:78 ILIRAP:97 CYP7A1:131 ABCD3:167
HSLAQ_LIVER_SPECIFIC_GENES	0.08571511	183	6.695e-05	3.854e-02	GYS2:34 MTPP:36 ALB:51 LGALS4:60 TMSF4:96 HP:90
REACTOME_BIOLOGICAL_OXIDATIONS	0.10142111	123	1.058e-04	4.952e-02	CYP11A1:4 PODXL2:66 MGST3:80 CYP7A1:131 COMT:180 CYP2F1:253
REACTOME_METABOLISM_OF_LIPIDS	0.04752273	581	1.071e-04	4.952e-02	CYP11A1:4 FAR2:18 PIP5K1A:25 EHHAH2:29 MORC2:30 ALB:51
REACTOME_SIGNALING_BY_WNT	-0.08640085	161	1.614e-04	6.967e-02	WNT4:25 DVL2:140 RAC2:161 H3C4:239 VPS26A:283 FRAT1:358
REACTOME_AMINO_ACID_TRANSPORT_ACROSS_THE	0.19410420	30	2.348e-04	9.504e-02	SLC6A12:119 SLC36A2:138 SLC6A15:194 SLC6A18:429 SLC6A6:478 SLC43A1:553
WP_Q213_COPY_NUMBER_VARIATION_SYNDROME	0.16072059	43	6.800e-04	1.021e-01	MERTK:70 IL1RL2:87 ILIRAP:97 ZC3H6:245 TTL:373 PSD4:380
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.20544963	26	2.889e-04	1.040e-01	CYP11A1:4 CYP7A1:131 COMT:180 HSD17B3:574 HSD17B8:679 SULT2B1:692
WP_OXIDATIVE_STRESS_RESPONSE	0.20027988	27	3.170e-04	1.081e-01	NOX1:213 TXNRD1:282 NQO1:374 MGST1:402 CYP1A1:695 HMOX1:923
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.10684806	92	4.049e-04	1.311e-01	CYP11A1:4 MGST3:80 CYP7A1:131 COMT:180 CYP2F1:253 GPX4:371
HOSHIDA_LIVER_CANCER_SUBCLASS_S3	0.06908213	210	5.853e-04	1.805e-01	SLC23A1:11 EHHAH2:29 SLC6A12:119 MMUT:175 AGXT:219 IVD:264
REACTOME_DIGESTION_OF_DIETARY_LIPID	0.48407682	4	7.991e-04	2.070e-01	CLPS:1 PNLP1:37 PNLP1RP1:336 PNLP1RP2:442 NA NA
BLALOCK_ALZHEIMERS_DISEASE_DN	-0.03617836	765	7.857e-04	2.070e-01	PTEN:5 ATP6V1:10 STXBPI:64 RALB:67 PCDH11X:69 NDUFAB8:88
CHIANG_LIVER_CANCER_SUBCLASS_PROLIFERATI	0.08421263	136	7.164e-04	2.070e-01	EHHADH:29 GYS2:34 MTPP:36 ABCB11:56 SLC22A1:62 GCDH:78
WOO_LIVER_CANCER_RECURRENCE_DN	0.12599568	60	7.449e-04	2.070e-01	MTPP:36 GCDH:78 SLC2A9:173 AASS:288 ACADS:56 SARDB:634
REACTOME_DISORDERS_OF_TRANSMEMBRANE_TRAN	0.08259665	133	1.031e-03	2.568e-01	SLC1A1:12 ABCB11:56 SLC36A2:138 NUP133:147 SLC2A9:173 SLC26A4:296
STARK_YYPOCAMPUF_22Q11_DELETION_DN	0.22304477	17	1.079e-03	2.589e-01	HIRA:61 TRMT2A:169 COMT:180 ARVCF:608 MRPL40:735 ZDHHC8:933
REACTOME_HEME_DEGRADATION	0.29579473	10	1.200e-03	2.658e-01	ALB:51 FABP1:179 HMOX1:923 UGT1A1:925 SLC01B1:1202 BLVRB:1531
REACTOME_BETA_CATENIN_INDEPENDENT_WNT_SI	-0.10463927	80	1.231e-03	2.658e-01	WNT4:25 DVL2:140 RAC2:161 FZD6:407 AGO3:414 RAC1:451
REACTOME_SENSORY_PERCEPTION_OF_TASTE	0.17407477	29	1.181e-03	2.658e-01	TAS2R41:71 TAS2R4:208 SCNN1G:358 TAS1R1:385 SCNN1B:406 TAS1R3:521
WP_FOCAL_ADHESION	-0.07403851	158	1.359e-03	2.825e-01	PTEN:5 BIRC3:58 CRK:158 RAC2:161 MAPK8:211 VEGFD:282
BERTUCCI_MEDULLARY_VS_DUCTAL_BREAST_CANC	-0.08427575	121	1.396e-03	2.825e-01	GUCY1A2:127 UACA:170 SNRNP:218 ADAMTS12:267 SPOCK1:286 ACTN1:4699.5
ANDERSEN_LIVER_CANCER_KRT19_DN	0.12370233	54	1.678e-03	2.841e-01	GNMT:174 PCYOX1:287 SELENBP1:414 PTGR1:540 HPD:760 AOX1:779
CARRILLOREIXACH_HEPATOBLASTOMA_VS_NORMAL	0.03351551	785	1.642e-03	2.841e-01	CYP11A1:4 HMCE5:6 SLC1A1:12 EHHAH2:29 GYS2:34 ABCB11:56
KEGG_TASTE_TRANSDUCTION	0.17611416	27	1.543e-03	2.841e-01	TAS2R41:71 TAS2R4:208 SCNN1G:358 TAS1R1:385 SCNN1B:406 ADCY4:460
REACTOME_TRANSLATION	-0.06703756	189	1.535e-03	2.841e-01	RPS8:32 MRPL39:78 SEC61B:106 MRPL37:139 FARSA:194 DAP3:213
REACTOME_DIGESTION_AND_ABSORPTION	0.23391898	15	1.711e-03	2.841e-01	CLPS:1 PNLP1:37 PNLP1RP1:336 PNLP1RP2:442 GUCA2A:558 SLC2C2:899
YOSHIMURA_MAPK8_TARGETS_UP	0.03201275	877	1.546e-03	2.841e-01	CLPS:1 SLC1A1:12 GYS2:34 PNLP1:37 RBP2:57 LGALS4:60
KEGG_FOCAL_ADHESION	-0.07267361	158	1.666e-03	2.841e-01	PTEN:5 BIRC3:58 CRK:158 RAC2:161 MAPK8:211 VEGFD:282
VERNOCHET_ADIPOGENESIS	0.24823034	13	1.944e-03	2.861e-01	HP:90 APCDD1:157 MC2R:224 CYP2F1:253 FFAR2:426 RARRES2:1197

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Kidney Calculi	0.17664677	51	1.042e-05	1.018e-01	PTH:77 CLDN16:103 SLC2A9:173 SLC13A2:205 AGXT:219 SLC4A1:312
Cystic Fibrosis	0.06158866	374	5.133e-05	1.671e-01	BPIFB1:2 SGSM3:46 ALB:51 NPC1:116 WHD88:125 NR1I2:133
Hyperkalemia	0.30300447	15	4.865e-05	1.671e-01	CYP11A1:4 SCNN1G:358 SCNN1B:406 SCNN1A:653 AC12:936 WNK1:975
Hypertensive disease	0.03945789	918	7.795e-05	1.903e-01	CYP11A1:4 ALB:51 TXNRD3:58 PTH:77 HP:90 PRKCD:91
Anemia, hereditary spherocytic hemolytic	0.39463988	5	2.243e-03	7.301e-01	SPTA1:33 SLC4A1:312 ANK1:604 EPB42:2260 SPTB:3398 NA
Hyperlipidemia, Familial Combined	0.11397288	62	1.941e-03	7.301e-01	MTPP:36 ALB:51 CYP7A1:131 FADS3:269 TXNIP:363 CUX1:610
Hypoxia-Ischemia, Brain	0.31688639	8	1.913e-03	7.301e-01	NOS2:197 TNF:1163 HIF1A:1560 IRAK1:1736 IL6:1791 VEGFA:1801
PSEUDOHYPALDOSTERONISM, TYPE IID	0.35584088	6	2.541e-03	7.301e-01	SLC12A3:356 WNK1:975 PRPF4B:1079 TRPV5:1660 WNK4:3177 NEK1:3567
22q11 partial monosomy syndrome	0.28010754	10	2.164e-03	7.301e-01	HIRA:61 COMT:180 ARVCF:608 UFD1:1138 GP1BB:2250 PI4KA:2339
Abnormality of aortic arch	0.27826471	10	2.314e-03	7.301e-01	HIRA:61 COMT:180 ARVCF:608 UFD1:1138 GP1BB:2250 MCTP2:2435
Benign Prostatic Hyperplasia	0.05466704	266	2.304e-03	7.301e-01	CYP11A1:4 GPRC5A:24 PROM1:40 TXNRD3:58 GYS2:34
Benign recurrent intrahepatic cholestasi	0.37988371	6	1.272e-03	7.301e-01	ABCB11:56 GGTLC3:793 GGT2:1616 ATP8B1:1886 GGT1:2045 GGTLC1:2577
Cardiovascular Diseases	0.03874618	561	1.999e-03	7.301e-01	MTPP:36 ALB:51 TXNRD3:58 PTH:77 HP:90 NPC1:116
Cholestasis	0.06710457	229	5.054e-04	7.301e-01	SLC23A1:11 ABCB11:56 SLC22A1:62 HP:90 NPC1:116 CYP7A1:131
Chronic Kidney Diseases	0.05686537	263	1.138e-03	7.301e-01	ALB:51 SLC22A1:62 PTH:77 HP:90 CLDN16:103 CHDH:127
Classical galactosemia	-0.25887603	14	7.995e-04	7.301e-01	GAL16 S100A1:21 GALK1:228 LGALS1:259 DMD:324 TGFBI:4699.5
Fatty Liver	0.05037213	312	2.409e-03	7.301e-01	SUGP1:27 MTPP:36 SGSM3:46 ALB:51 HP:90 NPC1:116
Hereditary spherocytosis	0.20908870	21	9.145e-04	7.301e-01	SPTA1:33 SLC4A1:312 ANK1:604 CAD:730 SLC26A8:768 G6PD:828
Hyperemia	0.19175945	25	6.241e-04	7.301e-01	PTH:77 NOS2:197 ADRB1:559 PTGS2:587 HRH1:606 SERPINE1:739
Hypocalcemia	0.13985742	40	2.228e-03	7.301e-01	HIRA:61 PTH:77 CLDN16:103 CHDH:127 COMT:180 SLC4A1:312
Impaired T cell function	0.23614517	15	1.547e-03	7.301e-01	HIRA:61 COMT:1